

EXPERIMENT NO. 13:

Implement the Car Price Prediction Model using Python

Aim:

To implement a Car Price Prediction Model in Python using Linear Regression and predict the price of a car based on its features.

Procedure:

1. Import the required Python libraries.
2. Create a sample car price dataset.
3. Separate the input features and target price.
4. Split the dataset into training and testing sets.
5. Create and train the Linear Regression model.
6. Predict car prices using the test data.
7. Calculate the model accuracy using R^2 score.
8. Predict the price of a new car.

Program:

Car Price Prediction using Linear Regression

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Sample Car Dataset
data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2017, 2019],
    'Kilometers': [80000, 70000, 60000, 50000, 40000,
                  30000, 20000, 15000, 55000, 35000],
    'Engine_CC': [1200, 1300, 1400, 1500, 1600,
                 1500, 1600, 1800, 1400, 1600],
    'Price': [350000, 400000, 450000, 520000, 600000,
             680000, 750000, 900000, 480000, 650000]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Input features and target
```

```
X = df[['Year', 'Kilometers', 'Engine_CC']]
```

```
y = df['Price']
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Create Linear Regression model
```

```
model = LinearRegression()
```

```
# Train the model
```

```
model.fit(X_train, y_train)
```

```
# Predict test data
```

```
y_pred = model.predict(X_test)
```

```
# Calculate R2 Score
```

```
accuracy = r2_score(y_test, y_pred)
```

```
print("Actual Prices:")
```

```
print(y_test.values)
```

```
print("\nPredicted Prices:")
```

```
print(y_pred.round(2))
```

```
print("\nR2 Score:")

print(round(accuracy, 2))


# Predict price of a new car
new_car = [[2020, 25000, 1500]]

prediction = model.predict(new_car)


print("\nPredicted Price of New Car:")

print(round(prediction[0], 2))
```

Output:

```
''' Actual Prices:
    [480000 400000]

    Predicted Prices:
    [459460.15 393149.1 ]

    R2 Score:
    0.85

    Predicted Price of New Car:
    670642.67
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Result:

The Car Price Prediction Model was successfully implemented in Python using Linear Regression. The model predicted car prices based on the year, kilometers driven, and engine capacity, and successfully estimated the price of a new car.